

ITP- Quiz(A)

Name:

Time: 30 mins

ERP:_____

Total: 20 Marks

Date: 12/3/2024

Q1. You are tasked with designing a C++ program that models a **library management system**. The program should use a struct to represent books and perform various operations on a collection of books.[10 marks]

Each book has the following attributes:

1. **Title** (string): The name of the book.
2. **Author** (string): The name of the author.
3. **ISBN** (string): A unique identifier for the book.
4. **Price** (float): The cost of the book.
5. **Stock** (int): The number of copies available in the library.

Your program should:

1. Create an array of struct to store information for up to 100 books.
2. Implement the following functionalities:
 - Add a new book to the library.
 - Search for a book by its ISBN and display its details.
 - List all books by a specific author.
 - Update the stock of a specific book by its ISBN.
 - Display all books that cost less than a given price.

Ans:

Q2. Write a recursive function `removeX()` that given a string, compute recursively a new string where all the 'x' characters have been removed. E.g., **[5 Marks]**

. `removeX("xaxb")` returns "ab"

. `removeX("abc")` returns "abc"

. `removeX("xx")` returns ""

Hint: All occurrences of 'x' in string `s` can be removed by removing 'x' from `_rst` position (if exists) and recursively removing 'x' from the rest.

Q3: Short Questions

- a. Explain how memory is allocated for a dynamically created 2D array in C++ using new. [**2 Marks**]

Ans:

- b. Modify the following code to make it allocate memory dynamically for a 2D array instead of using a static array: [**2 Marks**]

```
int arr[3][4];
```

Ans:

- c. Consider the following code: ,what will happen if you forget to deallocate arr[2]. [**1 Mark**]

```
int** arr = new int*[5];  
for (int i = 0; i < 5; i++) {  
    arr[i] = new int[10]; }  
Ans:
```